

COM Sync Suite CDK 4.02

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Developer Tools

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Agenda

- Introduction to Conduits
- COM Sync suite architecture
- COM Sync suite objects
- Developing a Conduit
- Registering conduit
- Next Version



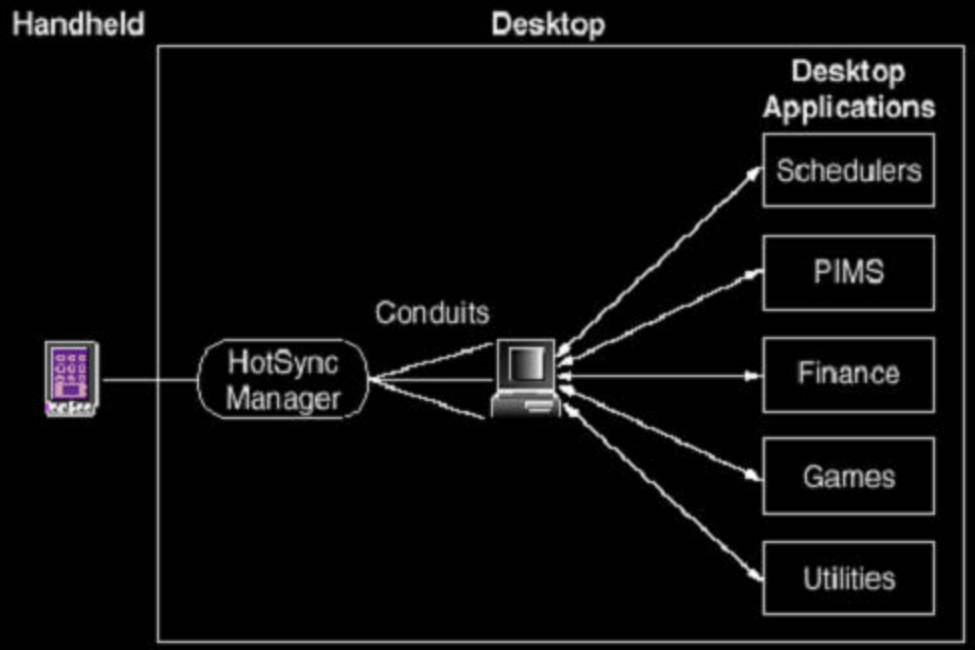
Introduction to Conduits

Using the Palm Powered™ handheld connected to a desktop, you can perform the following operations:

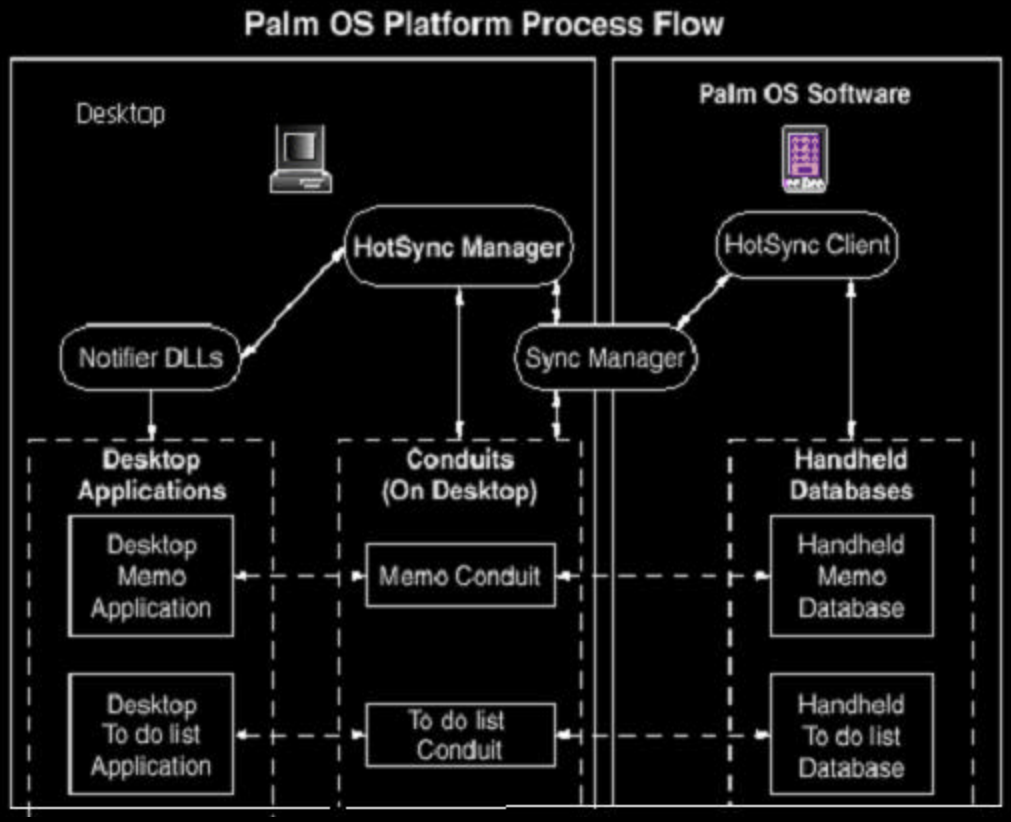
- synchronize data stored on the handheld with data stored on the desktop computer
- back up data stored on the handheld to the desktop computer
- install handheld applications that are stored on the desktop computer

Introduction to Conduits

Simplified View



Introduction to Conduits





Conduit Tasks

- Open and Close databases on the handheld
- Add, Delete and Modify records on the handheld and desktop computer, converting formats as required

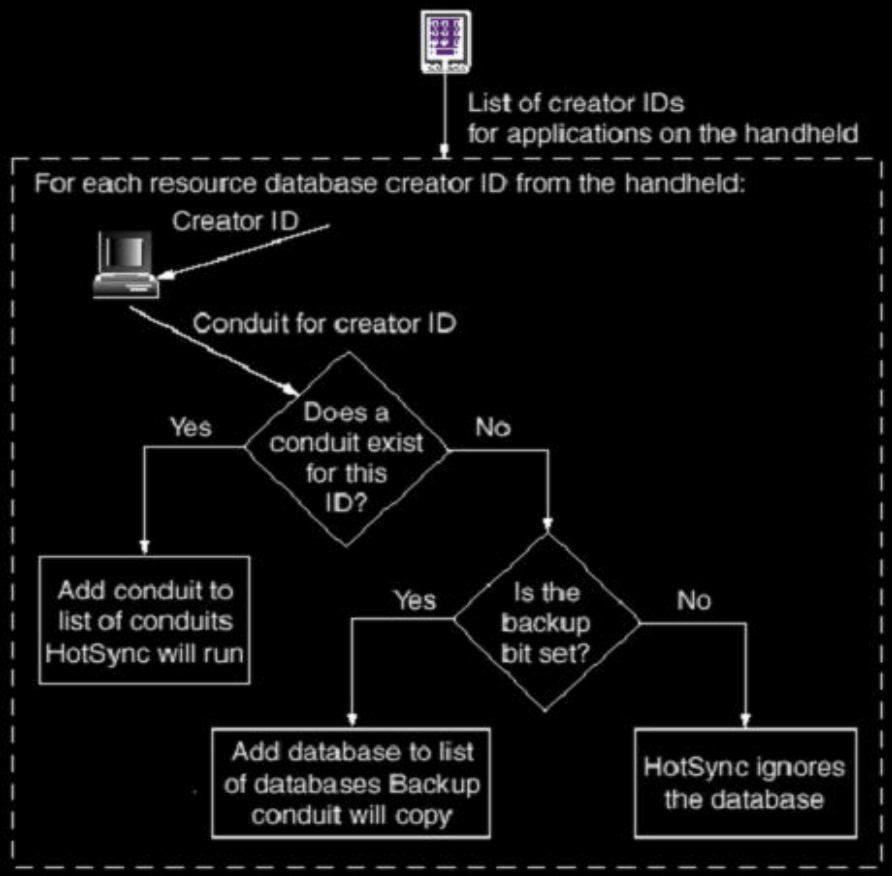


Standard Types of Synchronization

- Mirror image Sync
- One way Sync

CreatorID

Figure 4.2 How the HotSync Manager Determines Which Conduits to Run





CDK for Windows

The CDK for Windows consists of three suites:

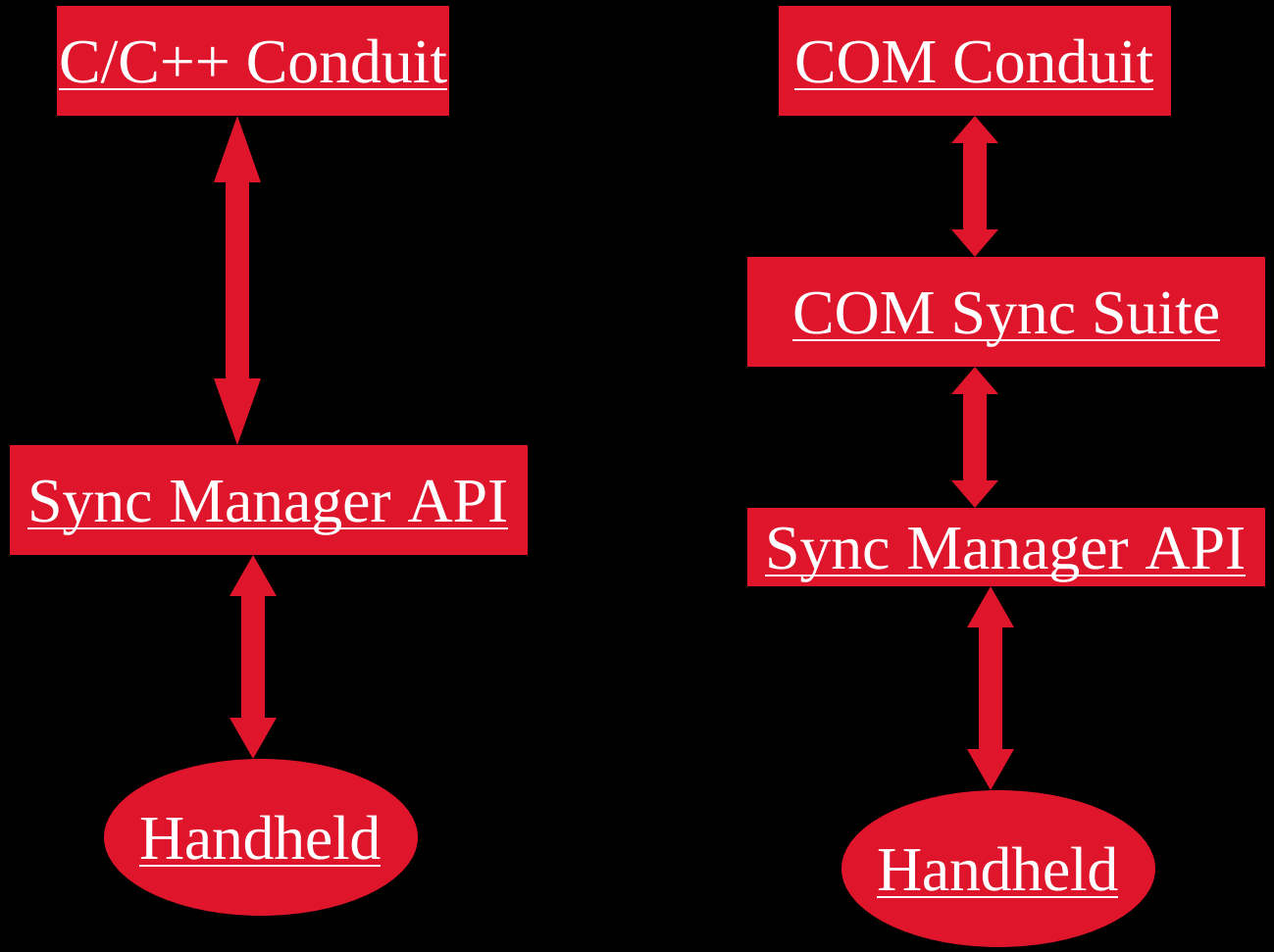
- **C/C++ Sync Suite**
- **COM Sync Suite**
- **JSync Suite**



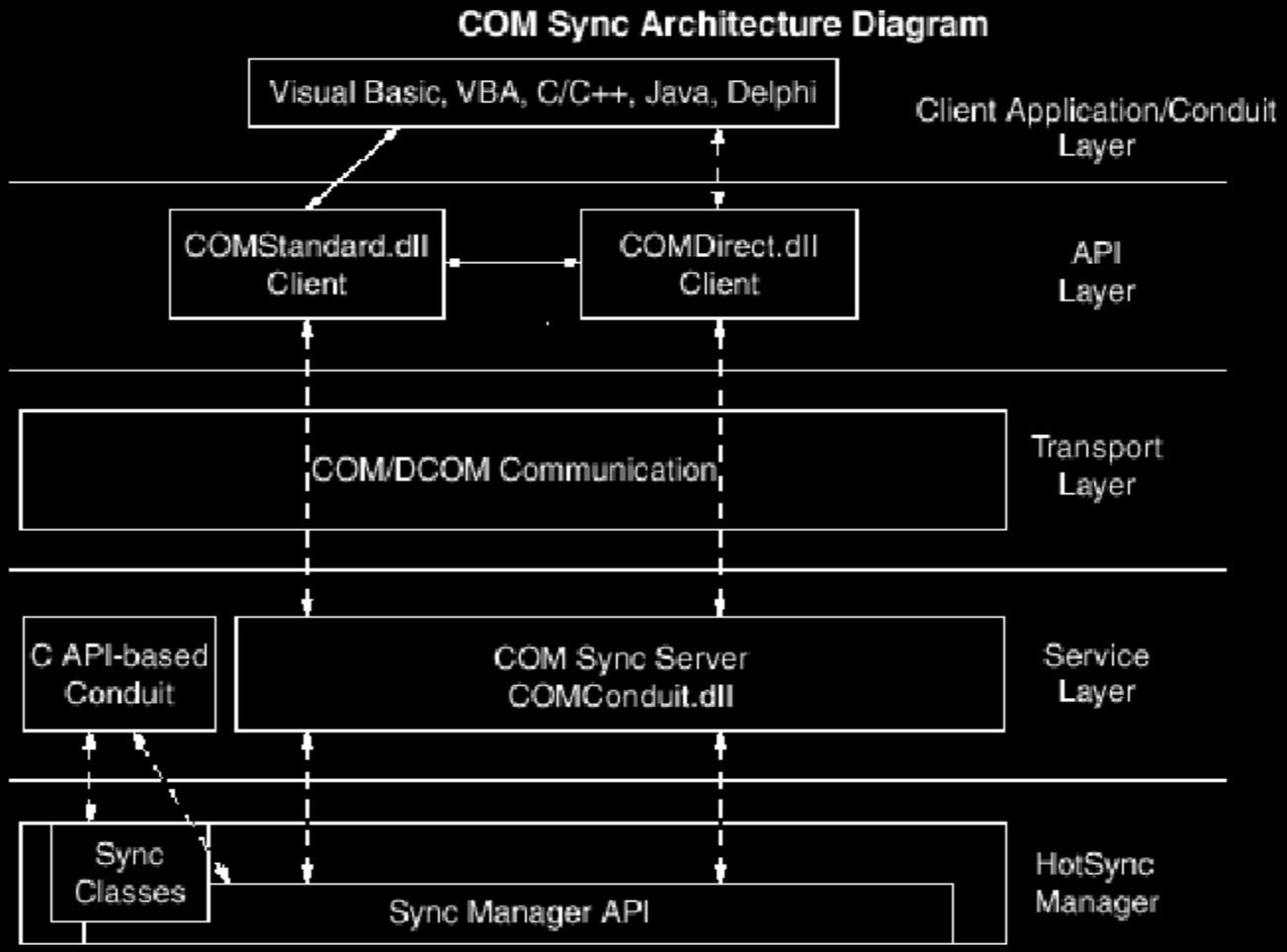
Introduction to COM Sync Suite

- Allows you to directly access Palm Powered™ handheld information from a desktop using any COM/DCOM-enabled programming language
- Environments includes Visual Basic , Borland C/C++, Delphi, Java, and Visual C/C++.

COM Sync Suite



Architecture





Features of COM Sync Suite

- You can debug your conduits using your IDE, keeping the handheld on-line. You have real-time access to your Palm data from your IDE, and the HotSync® Manager will not time-out.
- Lightweight Conduits -
  - written in C++/ATL without bulky MFC



Unique Features - I

- More than one application can talk to the handheld simultaneously



Unique Features - II

- You can open as many databases on the Palm Powered handheld as you want.

  You're no longer restricted to using only one database during synchronization.



Unique Features III

- You can write your own record adapter on top of our client, creating your own abstraction model.



Unique Features IV

- DCOM support

  - Ability to access a Palm OS® handheld across a network.

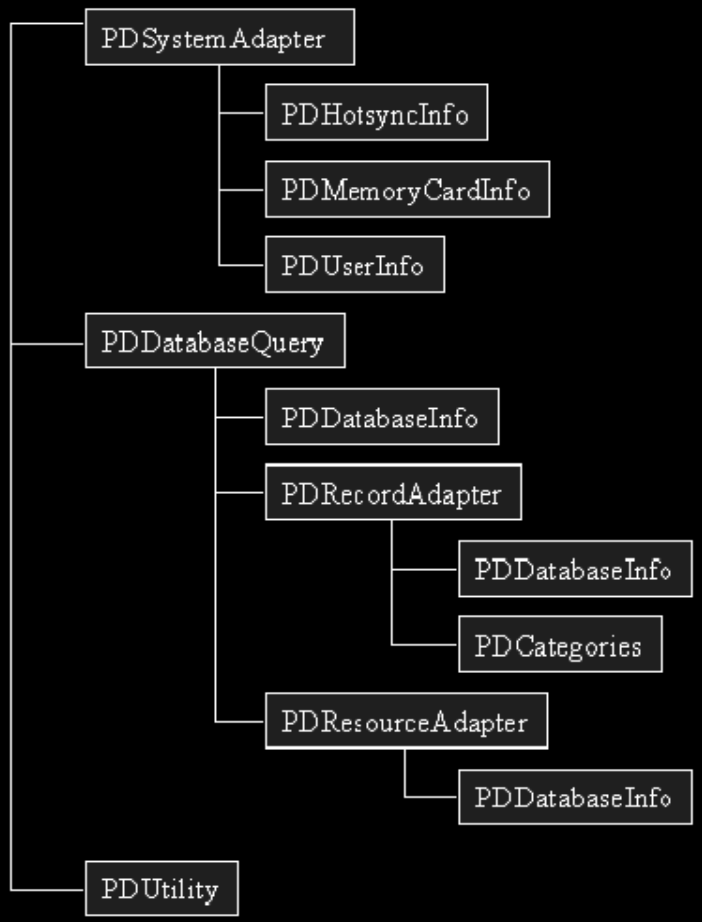


Unique Features - V

- Build conduits in multiple forms

  - Standard EXE
  - ActiveX EXE
  - ActiveX DLL

COM Sync Objects





COM Sync Objects

PDDatabaseQuery

- Represents the collection of databases in the handheld.
  - Allows to create/open databases
  - Provides list of databases available on HH
  - Reads database details.
  - Provides adapter objects for easier access to databases



COM Sync objects

- **PDDatabaseInfo**

  - Represents information about a handheld database.

  - provides details like name, creatorid, version, size, creation date, appinfo size, sortinfo size, record count etc.



COM Sync Objects

- **PDCategories**

  - provides access to database category information.

  - details like Name, ID, and Dirty.



COM Sync Objects

- **PDRecordAdapter**

  - represents an open record database.

  - read / write / modify records

  - iterator based or direct methods to access records

  - read / write appinfo block

  - read / write sortinfo block



COM Sync Objects

- **PDSystemAdapter**
  - **contains system info objects like**
    - **PDHotsyncInfo**
      - **information about the current HotSync session.**
    - **PDUserInfo**
      - **current handheld user information.**



COM Sync Objects

- **PDUtility**

  A collection of useful utility methods for manipulating strings, byte arrays, and integers.

  e.g., BSTRToByteArray

  ByteArrayToBSTR



IPDClientNotify

- Notification interface implemented by a COM based conduit client.
- Use standard EXE while developing the conduit
- Convert to ActiveX EXE/ActiveX DLL later



IPDClientNotify methods

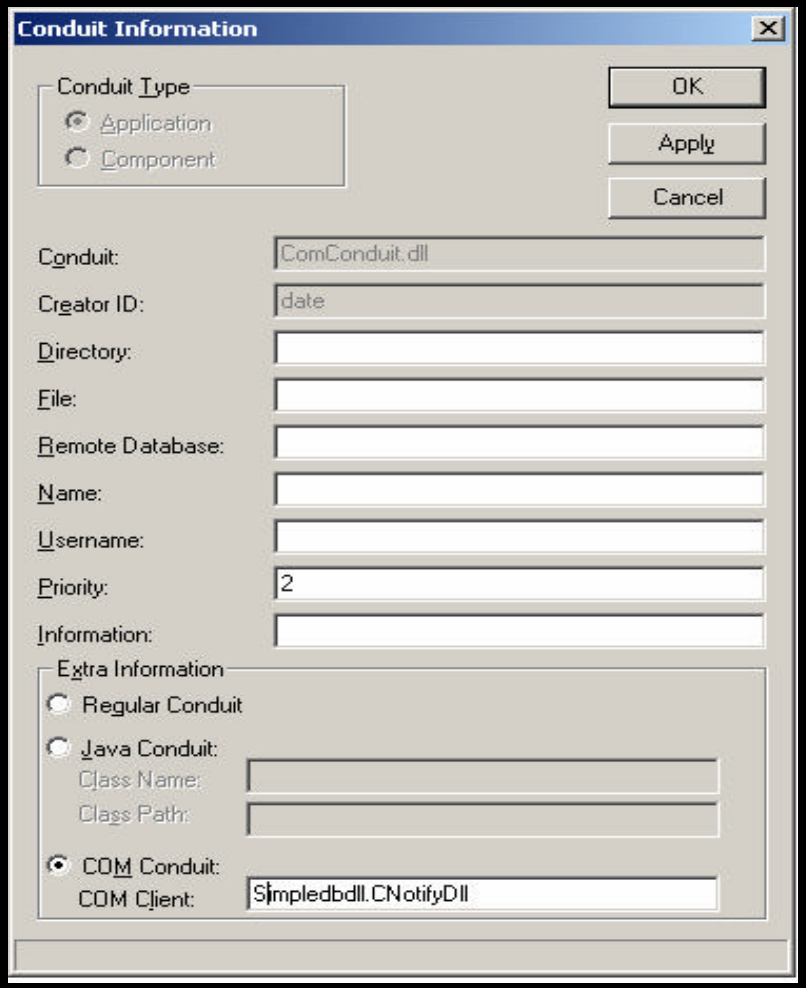
- BeginProcess
- CfgConduit
- GetConduitInfo



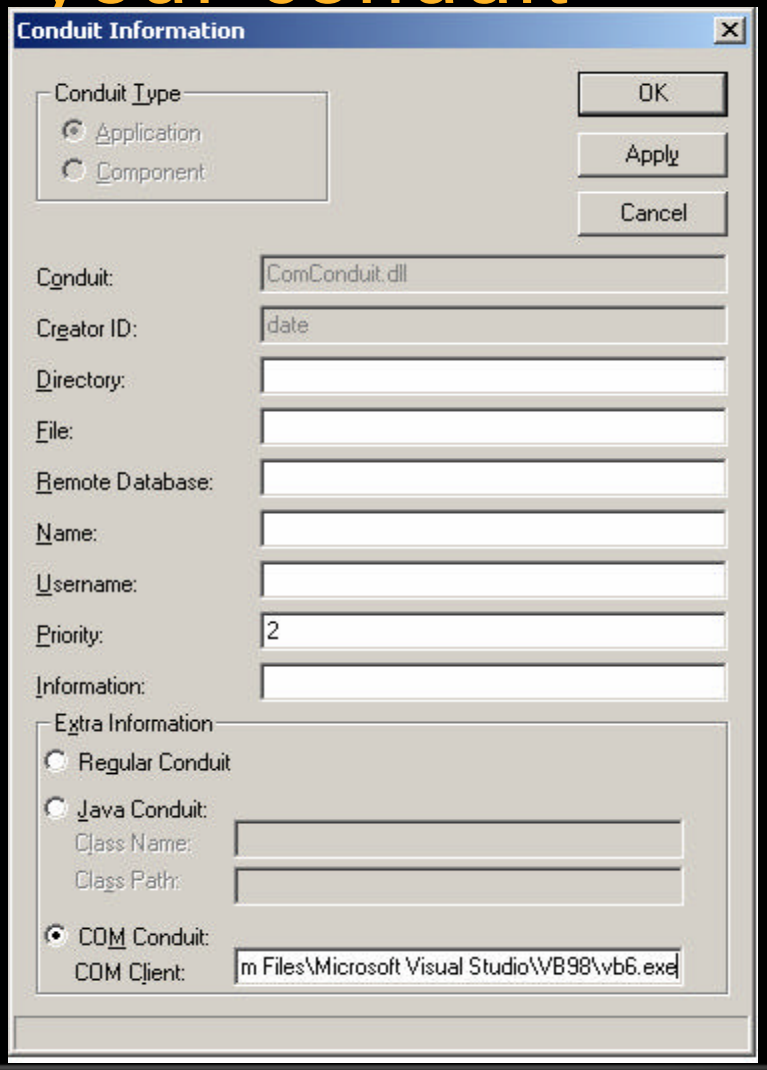
IPDClientNotify

- Developer must implement this interface to support configuration
- Use conduit registration to specify the object that implements the interface

Specify your object for Registration



Debugging your conduit





Upcoming enhancements

- Support for VisualBasic.Net / VisualStudio.Net
- Wizard for Visual C++.Net
- New PIM adapters
- New objects for Conduit registration, UserData
- New objects for Installing applications
- New objects for Expansion Manager



Questions/Answers

Questions/Answers



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